

Full-Stack Django + React Internship Preparation Notes

This PDF contains concise notes, interview preparation material, practice questions, and mini project guidance for mastering Django + React full-stack development for SDE internships.

Python Fundamentals

Core Concepts

- Variables, data types, loops, functions
- Object-Oriented Programming
- File handling and exception handling

Practice Questions

- 1 What is the difference between a list and tuple?
- 2 Explain inheritance in Python.
- 3 Write a function to reverse a string.

Example Code

```
def reverse_string(s):  
    return s[::-1]
```

JavaScript Essentials

Core Concepts

- DOM manipulation
- ES6 syntax
- Promises and async/await

Practice Questions

- 1 What is hoisting?
- 2 Difference between let, var, and const?
- 3 Explain async/await.

Example Code

```
async function fetchData() {  
    const res = await fetch('/api/data');  
    return await res.json();  
}
```

React Fundamentals

Core Concepts

- Components and JSX
- Props and state
- Hooks: useState and useEffect
- Routing with React Router

Practice Questions

- 1 What is the Virtual DOM?
- 2 Difference between props and state?
- 3 Why use useEffect?

Example Code

```
const [count, setCount] = useState(0);

useEffect(() => {
  console.log(count);
}, [count]);
```

Django Fundamentals

Core Concepts

- Models, Views, Templates
- Django ORM
- Authentication and CRUD

Practice Questions

- 1 What is Django ORM?
- 2 Explain migrations.
- 3 Difference between function-based and class-based views?

Example Code

```
class Task(models.Model):
    title = models.CharField(max_length=100)
    completed = models.BooleanField(default=False)
```

Django REST Framework

Core Concepts

- Serializers
- APIView and ViewSets
- JWT Authentication

Practice Questions

- 1 What is a serializer?

2 Why use JWT authentication?

3 What are ViewSets?

Example Code

```
class TaskSerializer(serializers.ModelSerializer):
    class Meta:
        model = Task
        fields = '__all__'
```

Full-Stack Integration

Core Concepts

- Connecting React with DRF APIs
- Authentication flow
- Deployment basics

Practice Questions

- 1 How does token authentication work?
- 2 What is CORS?
- 3 How do frontend and backend communicate?

Example Code

```
axios.get('/api/tasks/', {
  headers: {
    Authorization: `Bearer ${token}`
  }
});
```

Mini Projects

Project	Skills Covered
Task Manager	CRUD, JWT Auth, React Hooks
Blog Platform	Django Templates, Authentication
Expense Tracker	State Management, Charts
Job Portal	Advanced Full-Stack Architecture

Interview Preparation Checklist

- Build at least 2 deployed projects
- Practice 2–3 DSA problems daily
- Understand authentication and APIs
- Learn Git workflow and pull requests
- Prepare project explanations for interviews